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CIF – Cycle Invariant File

Shared Agronomic Framework for Crop Cycle Analysis

Supported by Innosuisse

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Background IP : Jan Affolter ITS (CHE-104.801.264), owner of the CompaniX project.

Foreword

Modern agriculture requires tools capable of describing agronomic reality with precision, without unnecessary complexity or interpretation. In a context where farming operations seek to understand performance gaps between seemingly similar plots, the ability to **document what the crop actually experienced** becomes essential.

The **CIF – Cycle Invariant File** addresses this need. It is a **standardized JSON format** designed to capture the **agronomic invariants** of a crop cycle, organized by **physiological stages**. The CIF does not model, predict, or simulate: it **observes, structures, and documents** the cycle exactly as it unfolded. It includes both the **conditions experienced by the crop** and the **management actions applied** throughout the cycle, ensuring a **complete and comparable representation** of agronomic reality.

The CIF integrates key agronomic elements such as **soil characteristics**, **varietal traits**, **growth indicators**, **management interventions**, and **yield summaries**. These components strengthen the **explanatory power** of the format while remaining **readable and directly usable** by agronomists, cooperatives, and institutions.

Detailed soil analyses are performed by specialized laboratories and incorporated into the CIF through the soil_analysis block, which provides a **structured and agronomically meaningful representation**. Raw laboratory datasets can remain in their respective systems, while the CIF retains the information necessary for **multi-plot comparison** and **agronomic interpretation**.

The CIF is designed to **empower farmers, agronomists, and agricultural authorities** in their decisions by providing a clear, structured view of the **conditions and actions that shaped crop performance**. Its purpose is primarily oriented toward **reporting, comparison, and decision-making**, rather than direct in-field operations. **However, in the CompaniX concept, the same sensors and field tools used to produce CIF invariants also support full autonomous precision agriculture, ensuring continuity between measurement and real-time operational decision-making.**

Within the CompaniX architecture, the CIF also provides the foundational layer for future cognitive intelligence capabilities. By aggregating and comparing large numbers of CIF datasets, these systems will progressively support advanced agronomic insights using empirical methods similar to those applied by agronomists.

Because the data can be available in real time, it enables farmers and advisors to go directly to the affected area to evaluate the situation and take action. This significantly increases the **efficiency of problem detection and treatment**, while reducing the risk of missing issues through random inspection or limited sampling.

This document presents the complete structure of the CIF and i**nvites agronomic feedback to reinforce its scientific consistency**. It is intended for cooperatives, technical institutes, agronomists, and advisors, and aims to establish **a shared methodological foundation** for the **objective analysis of crop performance**.

Table of Contents

1 Purpose of the Document....5
2 Role and Scope of the CIF....6
2.1 What the CIF Captures....6
a. Cycle Metadata, Global Climate, Soil, and Nutritional Invariants....6
b. variety_traits Block....6
c. soil_analysis Block....7
d. plant_invariants Block....7
2.2 Physiological and Operational Stages – stages[]....7
1. Pre-establishment....7
2. Establishment....7
3. Vegetative....8
4. Flowering....8
5. Grain Filling....8
6. Maturation....8
2.3 Yield Summary Blocks....9
2.4 What the CIF Does Not Capture....9
2.5 Role of the CIF within the CompaniX System....9
3 General Structure of the JSON....11
4 Why the CIF Represents a Major Advancement....11
5 Benefits of the CIF for Farmers....12
6 Benefits of the CIF for Agronomists....14
7 Benefits of the CIF for Agricultural Authorities....16
8 The Foundation of the Vision....18
9 The Vision....19
10 Toward a Global Cognitive Protocol for Agricultural Standardization....20
11 CompaniX Architecture....21
12 JSON CIF File....22
13 List of Acronyms Used....23
14 Project Ownership....24

1 Purpose of the Document

This document presents the JSON structure of the CIF – Cycle Invariant File and invites agronomic, nutritional, and systemic feedback to reinforce its scientific consistency and alignment with established field expertise.

The CIF is designed to be physiologically coherent, scientifically grounded, and sufficiently detailed to explain yield and quality differences between plots that share:

- the same soil type
- the same climate
- the same cultivated variety

Within this homogeneous context, the CIF enables the identification of the factors truly responsible for performance gaps: management, stress, variability, water dynamics, biotic pressure, physical or topographical constraints, **and the nutritional and systemic characteristics of the harvested crop**.

The CIF summarizes the crop cycle through agronomic invariants that reflect the conditions actually experienced by the crop, enriched by growth indicators (*growthmetrics*), nutritional and systemic invariants (*plant_invariants*), and summary blocks for yield and stress.

This document is addressed to cooperatives, agronomists, applied research institutes, food and feed professionals, and agricultural advisors, with the intention of fostering constructive dialogue and ensuring broad agronomic, nutritional, and systemic coherence.

2 Role and Scope of the CIF

The CIF (Cycle Invariant File) is the **documentary standard** used by CompaniX to describe, in a **structured and comparable manner,** the **agronomic, nutritional, and systemic conditions actually experienced** by a crop during a production cycle.

It is a **factual, neutral, and operational** file, designed to enable c**omparative performance analysis** between plots, between years, and between environments.

The CIF does not model the crop: it documents the **invariants**, **growth indicators**, **nutritional signatures**, and **yield summaries** that directly influence the **observed yield, quality, and nutritional value**.

2.1 What the CIF Captures

The CIF captures invariant and derived information that is:

- measurable
- comparable
- reproducible
- directly linked to crop performance, quality, and nutritional value

This information is organized at:

- the cycle level (plot identity, crop, variety, *variety_traits*, *soil_analysis*, *plant_invariants*, yield summaries)
- and across six physiological and operational stages: *pre_establishment* → *establishment* → *vegetative* → *flowering* → *grain_filling* → *maturation*

Each stage describes the key conditions and interventions of that specific phase.

a. Cycle Metadata, Global Climate, Soil, and Nutritional Invariants

Includes:

- identifiers (*cif_version*, *id_cycle*, *plot_id*)
- crop, variety, year
- sowing and harvest dates
- *season_climate_summary*: cumulative rainfall, GDD, ET0, and other cycle-level climate aggregates

b. variety_traits Block

Summarizes key varietal traits (earliness, drought tolerance, disease resistance, lodging resistance), distinguishing genetic potential from management or environmental effects.

c. soil_analysis Block

Consolidates essential laboratory soil information (texture, chemistry, macro- and micronutrients) into a standardized format, enabling comparison of plots based on objectified soil fertility and structure.

d. plant_invariants Block

Provides the nutritional and systemic signature of the harvested crop, including:

- NIR-measured composition (dry matter, protein, starch, NDF)
- derived nutritional values (digestibility, energy, amino acid pattern)
- systemic indicators (sovereignty, circularity, water pressure, nitrogen pressure, interannual stability)

This block enables the CIF to be used across human nutrition, food processing, livestock feeding, and circularity-oriented systems.

2.2 Physiological and Operational Stages – stages[]

The CIF now includes **six stages**, reflecting both operational and physiological phases of the crop cycle.

1. Pre-establishment

Captures the **conditions and operations before sowing**, including:

- tillage operations
- soil conditions (moisture, temperature, compaction)
- early weed pressure
- initial climate accumulations
- spatial variability (moisture CV, temperature CV, EC CV, microtopography)
- pre-sowing crop protection

This stage is essential for understanding:

- seedbed quality
- emergence potential
- early weed competition
- soil structure before planting

2. Establishment

Includes:

- early growthmetrics (LAI, NDVI, biomass, SPAD, thresholds)
- functional soil conditions (moisture, temperature, amplitudes)
- local climate (frost days, heat days, GDD, ET0)
- soil variability

- canopy microclimate (VPD, wetness events)
- management interventions (fertilization, crop protection)
- pathology indices

3. Vegetative

Captures:

- biomass accumulation
- canopy development
- nitrogen status
- soil moisture dynamics
- irrigation events
- disease pressure
- lodging risk
- microclimate variability

4. Flowering

A critical phase for yield potential, including:

- reproductive biomass
- heat stress indicators
- disease pressure
- canopy microclimate
- targeted fungicide applications

5. Grain Filling

Documents:

- grain loading dynamics
- late nitrogen status
- heat stress
- water stress
- soil moisture decline
- disease progression

6. Maturation

Captures:

- senescence
- final biomass
- late-season stress
- canopy decline
- harvest readiness indicators

2.3 Yield Summary Blocks

The CIF includes three complementary yield components:

1. yield_breakdown

Quantifies:

- potential yield
- losses by major limiting factors (water, nitrogen, diseases, frost, weeds, compaction/topography, others)
- actual yield

2. yield_summary

Provides:

- target yield
- actual yield
- harvest quality (moisture, protein, test weight)
- yield gap
- main limiting factor
- harvest index

3. yield

A direct, simple value of final yield.

4. constraints

A compact list of key limiting factors (e.g., *frost_at_establishment*, *water_stress_flowering*).

2.4 What the CIF Does Not Capture

The CIF does **not** contain raw time series (weather, sensors).

It contains only **agronomically and nutritionally relevant invariants and aggregates**, ensuring:

- compactness
- readability
- consistency across cycles
- integrity of laboratory and NIR data

2.5 Role of the CIF within the CompaniX System

The CIF is the **backbone** of CompaniX's agronomic, nutritional, and systemic analysis. **It enables:**

- rigorous comparison between crop cycles

- identification of explanatory factors for yield and quality gaps
- linking interventions, conditions, and outcomes
- building a coherent multi-year history
- integration with food and feed nutritional systems
- systemic evaluation (sovereignty, circularity, resource pressure)

The CIF is an **open standard**, designed to be **easy to populate**, **robust to analyze**, and **universally applicable** across agronomy, food systems, livestock nutrition, and circularity-oriented infrastructures.

3 General Structure of the JSON

The JSON describes a crop cycle through:

- **six stages** (pre_establishment → establishment → vegetative → flowering → grain_filling → maturation)
- invariants for soil, climate, variability, biotic stress, and management
- normalized indices (0–1), min/max values, amplitudes, CV
- a compact size of ~8–10 kB

A structure that is intentionally **simple, physiological, and comparable** across parcels and seasons.

4 Why the CIF Represents a Major Advancement

The CIF transforms agricultural traceability into:

- enriched agronomic data
- structured analysis by physiological stages
- reproducible multi-plot comparison
- official reporting
- strategic management insights
- **nutritional and systemic characterization of the harvested crop**

With its structured blocks (*soil_analysis*, *variety_traits*, *growthmetrics*, *yield_**, *plant_invariants*), the CIF provides a direct reading of:

- yield-limiting factors
- **nutritional outcomes (protein, starch, fiber, digestibility, energy)**
- **systemic performance (sovereignty, circularity, resource pressure, interannual stability)**

This enables diagnosis, improvement strategies, nutritional quality programs, and institutional reporting across agronomy, food systems, feed systems, and circularity-oriented infrastructures.

5 Benefits of the CIF for Farmers

The CIF can become the **single point of entry** for a farmer's operational traceability.
A single data entry, in one place, using the exact same information as usual.

CompaniX Automatically Generates Official Reports

From the CIF, CompaniX automatically produces:

- regulatory reports
- audit documents
- summaries requested by cooperatives
- standardized exports for agricultural authorities

→ The farmer enters data once.
→ CompaniX handles the rest.

A Clear Understanding of What Influenced Yield and Nutritional Quality

The CIF highlights:

- the alignment of the management schedule with climatic conditions
- timing variations between plots or years
- the actual reaction of the crop to soil and climate conditions
- the nutritional signature of the harvested crop (protein, starch, digestibility, energy)
- systemic indicators such as sovereignty, circularity, and resource pressure

If necessary, the CIF makes it possible to **adjust the management schedule** based on the conditions actually experienced in the field.

Objective Comparison Between Plots

Same soil, same variety, same weather… but not the same yield.
The CIF makes it possible to **understand why**.

Valuable Traceability

The CIF can be used for:

- quality assurance initiatives
- certifications
- audits
- administrative requests
- **food-sector quality programs (flour, starch, protein, processing suitability)**

- **feed-sector quality programs (digestibility, energy, amino acid patterns)**

Zero Additional Work

The system automatically enriches the data (active substance, mode of action, indices, etc.).
The farmer does **nothing more** than what they already do.

6 Benefits of the CIF for Agronomists

The CIF is designed to **empower field agronomists, advisors, and agricultural engineers**.
It transforms scattered observations into a **structured, comparable, and physiologically readable baseline**, improving both the **quality of diagnosis** and the **relevance of recommendations**.

A Structured Agronomic and Nutritional Reading of the Cycle

The CIF organizes information by **physiological and operational stages**, not simple calendar dates.
This allows agronomists to reason directly in relation to:

- crop physiology
- sensitive phases
- critical moments for yield formation
- **nutritional outcomes (protein, starch, fiber, digestibility, energy)**
- **systemic performance (sovereignty, circularity, resource pressure)**

Faster and More Objective Diagnostics

The CIF gathers in one place:

- conditions experienced by the crop
- stresses
- variability
- management interventions
- **nutritional composition and derived nutritional values**
- **systemic indicators**

This enables agronomists to **quickly identify explanatory factors** for yield gaps, using a more robust foundation than isolated observations.

Rigorous Comparison Between Plots and Seasons

The CIF facilitates comparisons between:

- seemingly similar plots
- successive seasons
- different management practices

It makes it possible to clearly distinguish what is driven by:

- soil
- climate
- stress
- variability
- management

Concrete Support for Recommendations

By linking interventions, their timing, their intensity, and the conditions actually experienced, the CIF helps agronomists formulate recommendations that are:

- more precise
- better argued
- better documented

Time Savings in Data Collection and Formalization

The CIF standardizes the information needed for agronomic analysis.
Agronomists spend **less time reconstructing a plot's history** and more time interpreting, comparing, and advising.

Support for Precision Agriculture and Quality Programs

CIF invariants can be fed by sensors and field tools also used for precision operations.
In real time, the agronomist can:

- target a specific zone
- verify it in the field
- decide more effectively on actions to take
- **anticipate nutritional impacts (protein, starch, digestibility)**

Direct View of Limiting Factors

The *yield_breakdown* and *yield_summary* blocks provide agronomists with a **direct reading of limiting factors**, without having to reconstruct the analysis from raw data.

7 Benefits of the CIF for Agricultural Authorities

The CIF constitutes a **standardized, neutral, and verifiable format** for documenting a crop cycle in a coherent, comparable, and scientifically sound manner.
It relies exclusively on data operators already report in their traceability registers, **without adding administrative burden**.

A Single Point of Entry for Operational Traceability

The CIF centralizes all mandatory data in a **structured and harmonized format**.

CompaniX Automatically Generates Regulatory Reports

From the CIF, CompaniX produces:

- formalized reports for authorities
- audit documents
- standardized exports
- summaries of farming practices

This guarantees **methodological consistency** between traceability, agronomic analysis, and reporting.

A Harmonized and Comparable Format

The CIF facilitates:

- regional or national consolidation
- practice analysis
- input tracking
- sustainability programs
- agroecological transition initiatives
- nutritional quality monitoring (protein, starch, energy, digestibility)
- systemic performance monitoring (sovereignty, circularity, resource pressure)

Enriched and Verifiable Traceability

Interventions are automatically enriched with:

- active substance
- mode of action
- formulation
- concentration
- timing and intensity indices
- nutritional invariants derived from NIR and validated equations
- systemic indicators supporting sustainability reporting

This reinforces the **reliability and transparency** of the data.

A Physiological Vision of the Cycle

The CIF structures data by **physiological and operational stages**, enabling a more relevant agronomic reading than simple calendar dates.

Compatibility with Public Policies

The CIF supports:

- input reduction strategies
- climate resilience programs
- certification processes
- practice monitoring systems

8 The Foundation of the Vision

The CIF is **not merely a measurement and reporting tool**: it constitutes the **field-level building block of a broader cognitive infrastructure**.

The "Vision" chapter describes how this infrastructure brings forth a **global cognitive protocol for agricultural standardization**, built on **collective learning** and **territorial sovereignty**.
By reading it, you will see how the data structured by the CIF becomes the **foundation of a system** capable of connecting farms, territories, and institutions within a **common agronomic language** – a **Global Agricultural Standardization Protocol**.

9 The Vision

A vision that opens a possible future

There are projects that do more than provide a solution. They reveal a possible future. They show what a system could become if we choose to make it evolve rather than endure it. This document belongs to that rare category.

It does not merely describe an architecture or a protocol. It outlines the contours of a world where agriculture, health, and nutrition no longer operate in silos, but as a living organism — capable of learning, adapting, and protecting the populations it serves.

As you read these pages, you will discover a system that connects farms to one another, that lets field data converse with market signals, and that transforms every local observation into collective knowledge. What you are about to read is not an abstraction: it is a concrete, possible trajectory, already in motion.

Imagine…

Imagine an agriculture where every decision — sowing, irrigating, treating, harvesting — enriches a collective intelligence that improves with every cycle. An agriculture where soil data, climate data, yields, markets, and even the nutritional and health needs of populations no longer remain isolated. They become living knowledge — shared, connected, immediately actionable.

Imagine a system where every farm progresses thanks to the others, where every territory gains resilience, and where every data point strengthens collective performance. A system capable of anticipating risks, optimizing resources, and aligning agricultural production with real nutritional and health needs.

This is the vision carried by CompaniX.

A project that does not seek to replace what exists, but to reinforce it. A project that offers a path to secure food resources, support farmers, anticipate crises, and align production with the real needs of populations.

Some projects are important. Others are necessary. And then there are those that, by their very nature, call to be understood, supported, and extended — because they touch what underpins the stability of a society: its ability to nourish, protect, and anticipate.

CompaniX is one of those projects.

10 Toward a Global Cognitive Protocol for Agricultural Standardization

An Emerging Global Protocol

CompaniX is not limited to optimizing farms or aggregating data. By orchestrating distributed technologies, evolving ontologies, sovereign knowledge graphs, and an antifragile cognitive learning system powered by frugal cycle capture and field tools (IoT sensors, robots, drones used both for measurement and operational execution), the system gradually creates a **shared language** between farms, regions, financial institutions, and actors across the food chain.

This shared language — built from definitions, invariants, cycles, responses to shocks, and operational standards — becomes, in practice, a **reference protocol**. It enables different regions to compare their situations **without sharing their data**, to align their practices **without standardizing their contexts**, and to synchronize their decisions **while preserving full sovereignty**.

As the network expands, this protocol acquires a rare property: it becomes **self-reinforcing**. Each new region enriches the global knowledge base, each cycle improves the precision of recommendations, and each actor progressively adopts the same standards of measurement, evaluation, and remediation.

Without ever imposing it, CompaniX creates the conditions for the emergence of a **global protocol for agricultural standardization**, grounded in sovereignty, interoperability, and collective learning. A protocol capable of unifying practices, reducing systemic risks, and aligning agricultural production with real nutritional and health needs.

This is not a platform. It is a **cognitive infrastructure** from which a global standard emerges.

A Lean Enterprise by Design

CompaniX was conceived from the outset to be **extremely lean** in every dimension. Operations rely on standardized, reproducible processes; the technological architecture is distributed and modular; and each region operates its own instance without central overhead. Data does not circulate: only **learning** propagates, reducing costs, risks, and regulatory constraints.

Governance remains lightweight, based on standards rather than complex structures, and development follows a modular, iterative logic focused on **direct value for farms and territories**.

This (extremely) lean approach is not an aesthetic choice: it is a **structural requirement** for a global protocol to emerge, spread, and remain sovereign, effective, and economically viable.

11 CompaniX Architecture

```
==========================================================================================
                        COMPANIX ARCHITECTURE: STRICT ROLE SEPARATION
==========================================================================================


     [ ON-FARM / LOCAL ENVIRONMENT ]                        [ REMOTE / NATIONAL OR GLOBAL ]

   +-------------------------------------+                +-------------------------------------+
   |             EDGE NODE               |                |          COGNITIVE CORE             |
   |-------------------------------------|                |-------------------------------------|
   |                                     |                |                                     |
   |  [Sensors & IoT Devices]            |                |  [Data Ingestion Pipeline]          |
   |        | (Raw Data)                 |                |        |                            |
   |        v                            |                |        v                            |
   |  (Data Aggregation)                 |    Transmits   |  (Ontologies & Vectorization)       |
   |        |                            |     CIF 3.0    |        |                            |
   |        +--> Real-Time Management    | ===============>|        +--> Cross-Learning          |
   |        +--> Precision Operations    |  (Standardized |        +--> Yield Optimization      |
   |        +--> Instant Alerts          |   JSON File)   |        +--> Best Practices ID       |
   |        |                            |                |        +--> Official Reporting      |
   |        v                            |                |                                     |
   |  [CIF Generation Engine]            |                |                                     |
   |                                     |                |                                     |
   +-------------------------------------+                +-------------------------------------+
         ^                            ^                                |                  |
         |                            |      Collective Knowledge,     |                  |
         |                            |    Insights & Formal Reports   |                  |
         |                            +----------------+-----------------+                  |
         |                                             |                                    |
         v                                             v                                    v
+-----------------------+              +---------------------------+             +-----------------------+
|      THE FARMER       |              | AGRICULTURAL AUTHORITIES  |             |    THE AGRONOMISTS    |
|  (Local Execution &   |              |  (Regulatory Compliance   |             | (Strategic Insights & |
|   Automated Reports)  |              |   & Regional Monitoring)  |             |  Yield Optimization)  |
+-----------------------+              +---------------------------+             +-----------------------+
```

12 JSON CIF File

The CIF file has been removed from this document and is now available on GitHub :

https://github.com/janaffolter/CompaniX-CIF/blob/main/cif.json

13 List of Acronyms Used

Acronym	Meaning	Description
CIF	Cycle Invariant File	CompaniX JSON file representing a crop cycle through agronomic invariants.
CV	Coefficient of Variation	Measure of spatial variability (relative differences in moisture, temperature, EC, etc.).
EC	Electrical Conductivity	Indicator of soil salinity, structure, texture, and spatial variability.
VPD	Vapor Pressure Deficit	Key indicator of water stress and transpiration.
N-1	Previous Crop Cycle	Represented by the CIF from the previous season, capturing agronomic legacy.

14 Project Ownership

Supported by: Innosuisse

Owner & Founder: Jan Affolter — jan.affolter@gmail.com

Intellectual Property Holder: Jan Affolter ITS (CHE-104.801.264)

Location: Lausanne, Switzerland